



Chapter 10

E-commerce: Digital Markets, Digital Goods

Video Cases

Video Case 1: Deals Galore at Groupon

Video Case 2: Etsy: A Marketplace and Community

Video Case 3: Ford Manufacturing Supply Chain: B2B Marketplace



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LEARNING OBJECTIVES

- **What are the unique features of e-commerce, digital markets, and digital goods?**
- **What are the principal e-commerce business and revenue models?**
- **How has e-commerce transformed marketing?**
- **How has e-commerce affected business-to-business transactions?**
- **What is the role of m-commerce in business and what are the most important m-commerce applications?**
- **What issues must be addressed when building an e-commerce presence?**



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Otto Group: Profits from Prediction

- **Problem:** How to derive profits from large and varied consumer base
- **Solution:**
 - Create a data warehouse for customer data
 - Use **NeuroBayes** to analyze and discover patterns in data
- Demonstrates use of technology to increase efficiency and reduce costs



Unique Features of E-commerce, Digital Markets, and Digital Goods

- **E-commerce: Use of the Internet and Web to transact business.**
- **Began in 1995 and grew exponentially; still stable even in a recession.**
- **Companies that survived the dot-com bubble now thrive.**
- **The new e-commerce: social, mobile, local**



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The Growth of E-commerce

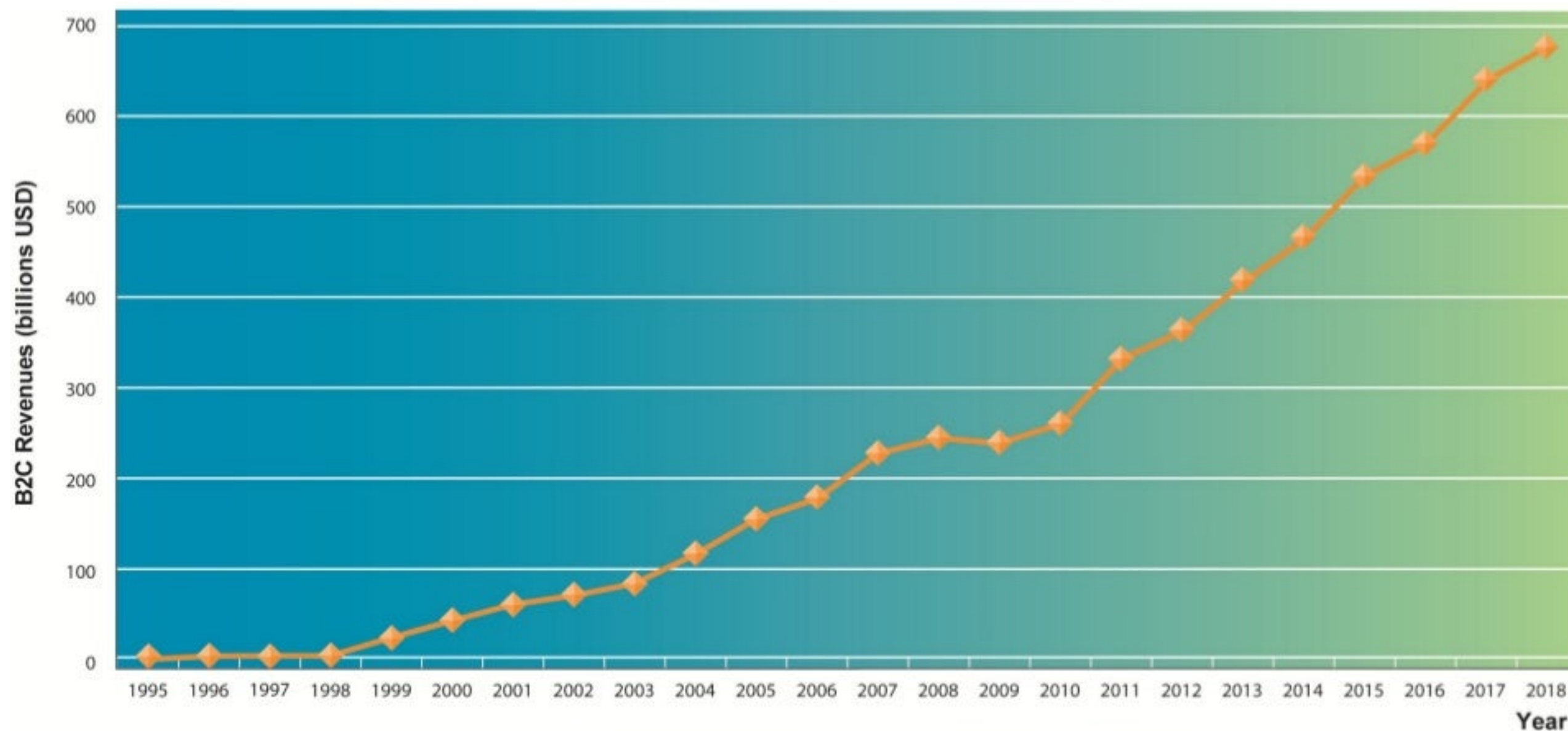


Figure 10-1

Retail e-commerce revenues grew 15–25 percent per year until the recession of 2008–2009, when they slowed measurably. In 2014, e-commerce revenues are growing again at an estimated 12 percent annually.



Unique Features of E-commerce, Digital Markets, and Digital Goods

- **Eight unique features of Internet and Web as commercial medium**
 - Ubiquity
 - Global reach
 - Universal standards
 - Richness
 - Interactivity
 - Information density
 - Personalization/customization
 - Social technology



Unique Features of E-commerce, Digital Markets, and Digital Goods

– Ubiquity

- Internet/Web technology available everywhere: work, home, and so on, anytime
- Effect:
 - Marketplace removed from temporal, geographic locations to become “marketspace”
 - Enhanced customer convenience and reduced shopping costs
- Reduces transaction costs
 - Costs of participating in market



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Unique Features of E-commerce, Digital Markets, and Digital Goods

- **Global reach**
 - The technology reaches across national boundaries, around Earth
 - Effect:
 - Commerce enabled across cultural and national boundaries seamlessly and without modification.
 - Marketspace includes, potentially, billions of consumers and millions of businesses worldwide.



Unique Features of E-commerce, Digital Markets, and Digital Goods

- **Universal standards**
 - **One set of technology standards: Internet standards**
 - **Effect:**
 - Disparate computer systems easily communicate with one another
 - Lower market entry costs—costs merchants must pay to bring goods to market
 - Lower consumers' search costs—effort required to find suitable products



- **Richness**

- **Supports video, audio, and text messages**
- **Effect:**
 - Possible to deliver rich messages with text, audio, and video simultaneously to large numbers of people.
 - Video, audio, and text marketing messages can be integrated into single marketing message and consumer experience.



- **Interactivity**

- The technology works through interaction with the user.
- **Effect:**
 - Consumers engaged in dialog that dynamically adjusts experience to the individual.
 - Consumer becomes co-participant in process of delivering goods to market.



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Unique Features of E-commerce, Digital Markets, and Digital Goods

- **Information density**
 - Large increases in information density—the total amount and quality of information available to all market participants
 - Effect:
 - Greater price transparency
 - Greater cost transparency
 - Enables merchants to engage in price discrimination



- **Personalization/Customization**
 - Technology permits modification of messages, goods
 - Effect:
 - Personalized messages can be sent to individuals as well as groups.
 - Products and services can be customized to individual preferences.



- **Social technology**
 - **The technology promotes user content generation and social networking**
 - **Effect:**
 - New Internet social and business models enable user content creation and distribution, support social networks
 - Many-to-many model



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Unique Features of E-commerce, Digital Markets, and Digital Goods

- **Effect of the Internet on the marketplace:**
 - Reduces information asymmetry
 - **Offers greater flexibility and efficiency because of:**
 - Reduced search costs and transaction costs
 - Lower menu costs
 - Greater price discrimination
 - Dynamic pricing
 - **May reduce or increase switching costs**
 - **May delay gratification: effects dependent on product**
 - **Increased market segmentation**
 - **Stronger network effects**
 - **More disintermediation**



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The Benefits of Disintermediation to the Consumer

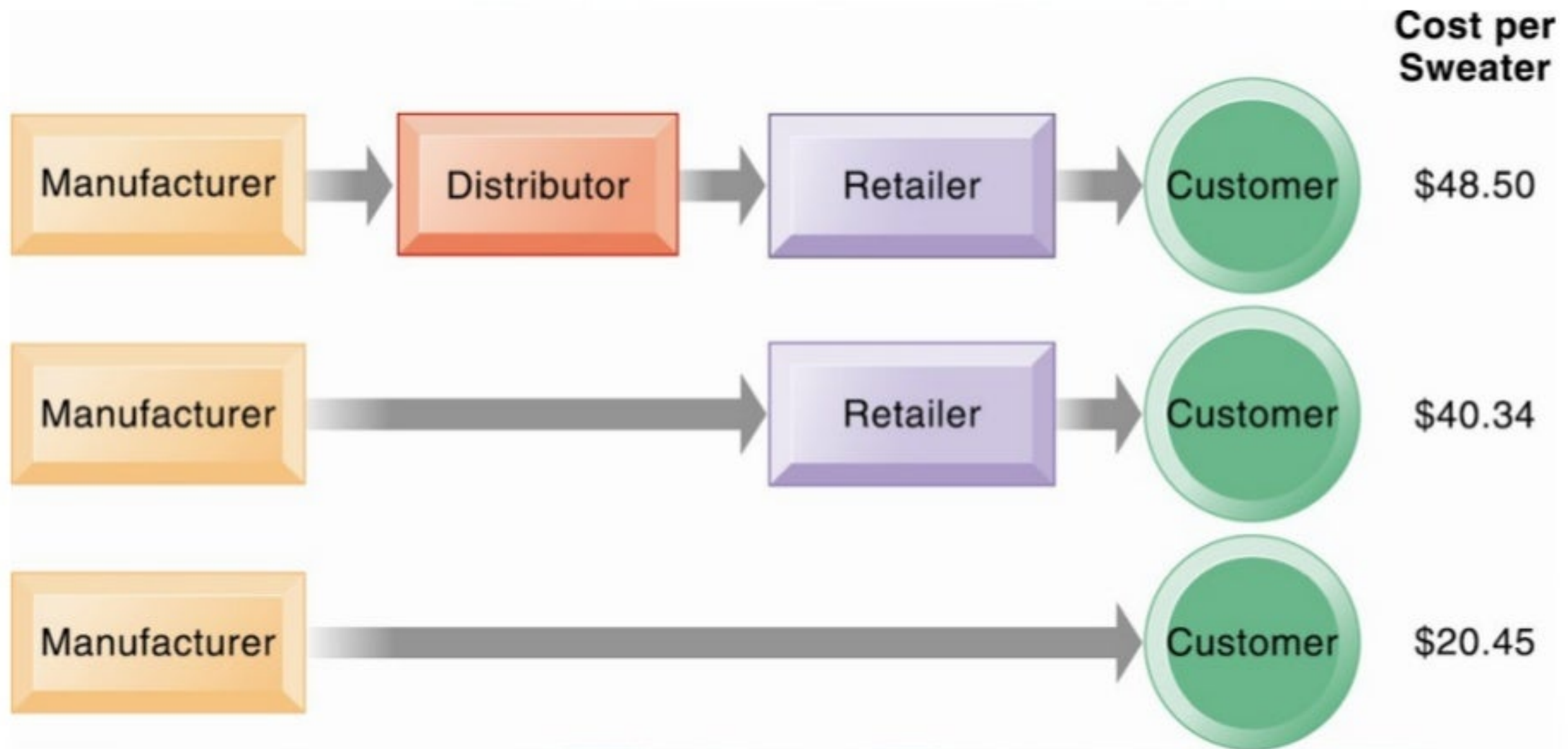


Figure 10-2

The typical distribution channel has several intermediary layers, each of which adds to the final cost of a product, such as a sweater. Removing layers lowers the final cost to the consumer.



Unique Features of E-commerce, Digital Markets, and Digital Goods

- **Digital goods**
 - **Goods that can be delivered over a digital network**
 - For example: music tracks, video, software, newspapers, books
 - **Cost of producing first unit is almost entire cost of product**
 - **Costs of delivery over the Internet very low**
 - **Marketing costs remain the same; pricing highly variable**
 - **Industries with digital goods are undergoing revolutionary changes (publishers, record labels, etc.)**



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E-commerce Business and Revenue Models

- **Three major types of e-commerce**
 - **Business-to-consumer (B2C)**
 - Example: BarnesandNoble.com
 - **Business-to-business (B2B)**
 - Example: ChemConnect
 - **Consumer-to-consumer (C2C)**
 - Example: eBay
- **E-commerce can be categorized by platform**
 - **Mobile commerce (m-commerce)**



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E-commerce Business and Revenue Models

- **E-commerce business models**
 - Portal
 - E-tailer
 - Content provider
 - Transaction broker
 - Market creator
 - Service provider
 - Community provider



- **E-commerce revenue models**
 - Advertising
 - Sales
 - Subscription
 - Free/Freemium
 - Transaction fee
 - Affiliate



Interactive Session: Organizations

Can Pandora Succeed with Freemium?

Read the Interactive Session and discuss the following questions

- **Analyze Pandora using the value chain and competitive forces models. What competitive forces does the company have to deal with? What is its customer value proposition?**
- **Explain how Pandora's "freemium" business model works. How does the company generate revenue?**
- **Can Pandora succeed with its "freemium" model? Why or why not? What people, organization, and technology factors affect its success with this business model?**



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How Has E-commerce Transformed Marketing?

- **E-commerce marketing**
 - Internet provides new ways to identify and communicate with customers.
 - Long tail marketing:
 - Ability to reach a large audience inexpensively
 - Internet advertising formats
 - Behavioral targeting:
 - Tracking online behavior of individuals on thousands of Web sites and within apps
 - Privacy concerns



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Web Site Visitor Tracking

E-commerce Web sites have tools to track a shopper's every step through an online store. Close examination of customer behavior at a Web site selling women's clothing shows what the store might learn at each step and what actions it could take to increase sales.

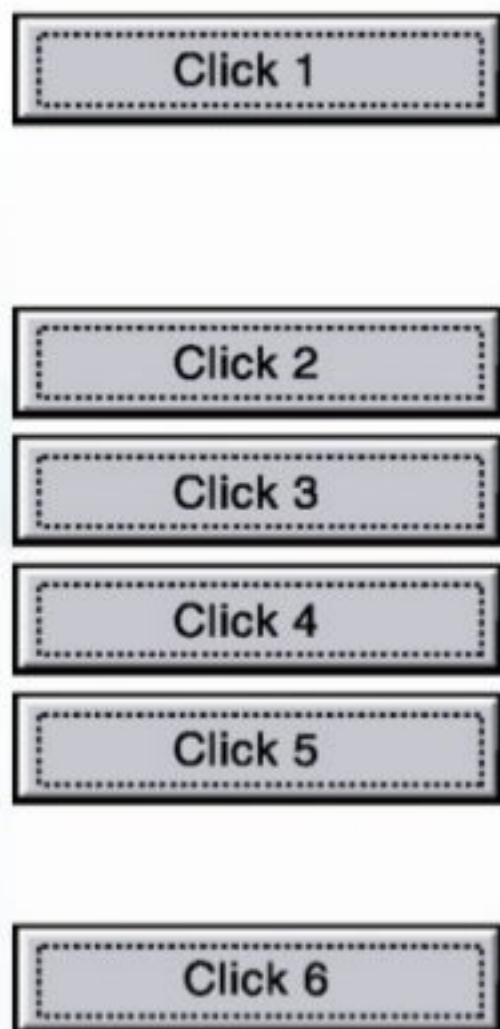


Figure 10-3

The shopper clicks on the home page. The store can tell that the shopper arrived from the Yahoo! portal at 2:30 PM (which might help determine staffing for customer service centers) and how long she lingered on the home page (which might indicate trouble navigating the site). Tracking beacons load cookies on the shopper's browser to follow her across the Web.

The shopper clicks on blouses, clicks to select a woman's white blouse, then clicks to view the same item in pink. The shopper clicks to select this item in a size 10 in pink and clicks to place it in her shopping cart. This information can help the store determine which sizes and colors are most popular. If the visitor moves to a different site, ads for pink blouses will appear from the same or different vendor.

From the shopping cart page, the shopper clicks to close the browser to leave the Web site without purchasing the blouse. This action could indicate the shopper changed her mind or that she had a problem with the Web site's checkout and payment process. Such behavior might signal that the Web site was not well designed.



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Web Site Personalization

Firms can create unique personalized Web pages that display content or ads for products or services of special interest to individual users, improving the customer experience and creating additional value.

Figure 10-4





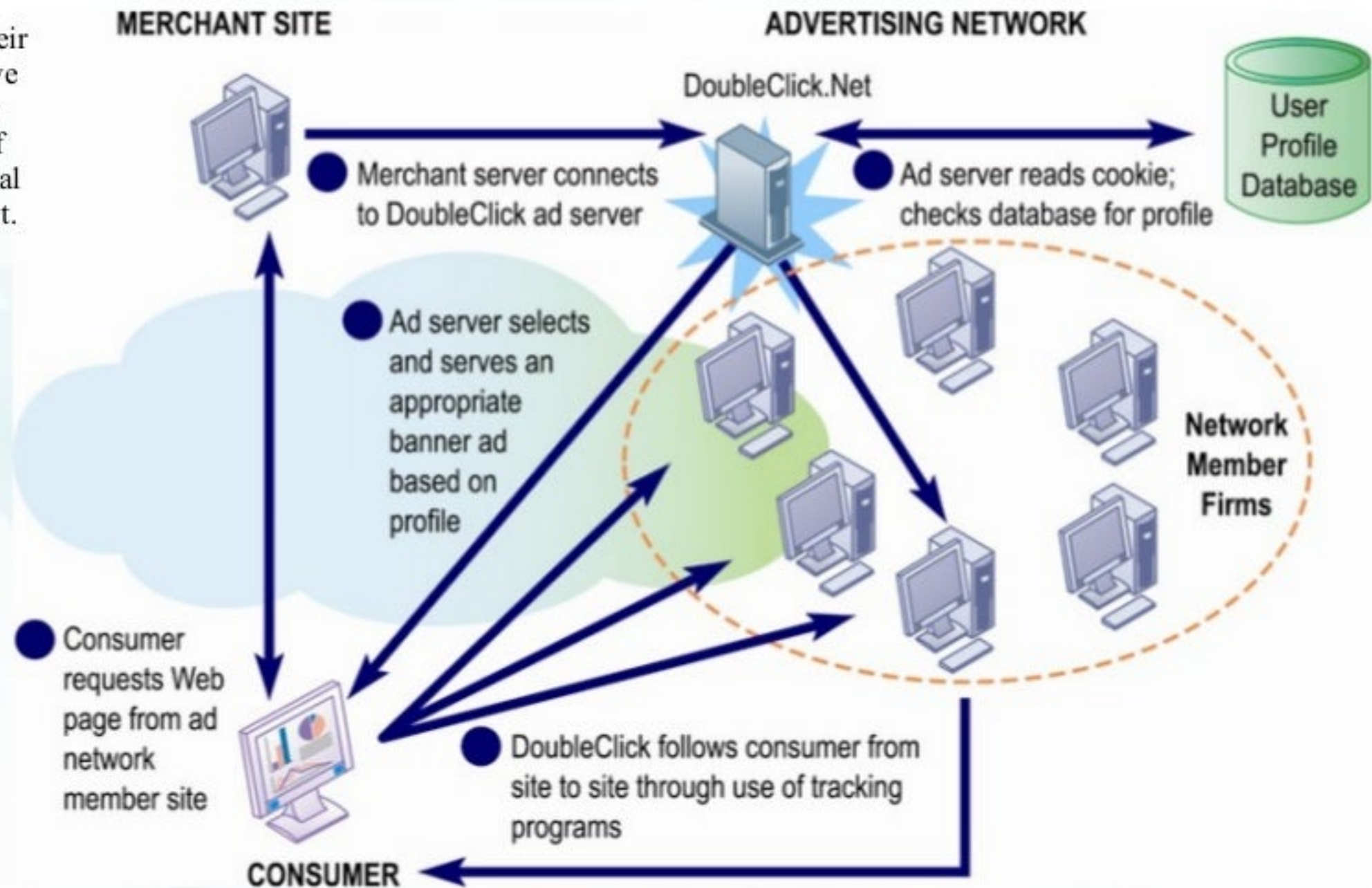
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How an Advertising Network Works

Advertising networks and their use of tracking programs have become controversial among privacy advocates because of their ability to track individual consumers across the Internet.

Figure 10-5





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E-commerce Business and Revenue Models

- **Social e-commerce:**
 - Based on digital social graph
- **Features of social e-commerce driving its growth**
 - Newsfeed
 - Timelines
 - Social sign-on
 - Collaborative shopping
 - Network notification
 - Social search (recommendations)



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E-commerce Business and Revenue Models

- **Social media:**
 - Fastest growing media for branding and marketing
- **Social network marketing:**
 - Seeks to leverage individuals influence over others in social graph
 - The target is a social network of people sharing interests and advice
 - Facebook's "Like button"
 - Social networks have huge audiences
 - Facebook: 137 million U.S. visitors monthly



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How Has E-commerce Transformed Marketing?

- **Social shopping sites**
- **Wisdom of crowds**
- **Crowdsourcing**
 - Large numbers of people can make better decisions about topics and products than a single person.
- **Prediction markets**
 - Peer-to-peer betting markets on specific outcomes (elections, sales figures, designs for new products)



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E-commerce and Business-to-Business Transactions

- **B2B e-commerce**
 - U.S. B2B trade in 2014 is \$13.8 trillion
 - U.S. B2B e-commerce in 2014 is \$5.7 trillion
 - Procurement requires significant overhead costs, which Internet and networking helps automate
 - Variety of Internet-enabled technologies used in B2B
 - Electronic data interchange (EDI)
 - Private industrial networks (private exchanges)
 - Net marketplaces
 - Exchanges



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E-commerce and Business-to-Business Transactions

- **Electronic data interchange (EDI)**
 - Computer-to-computer exchange of standard transactions such as invoices, purchase orders.
 - Major industries have EDI standards that define structure and information fields of electronic documents.
 - More companies are increasingly moving toward private networks that allow them to link to a wider variety of firms than EDI allows and share a wider range of information in a single system.



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Electronic Data Interchange

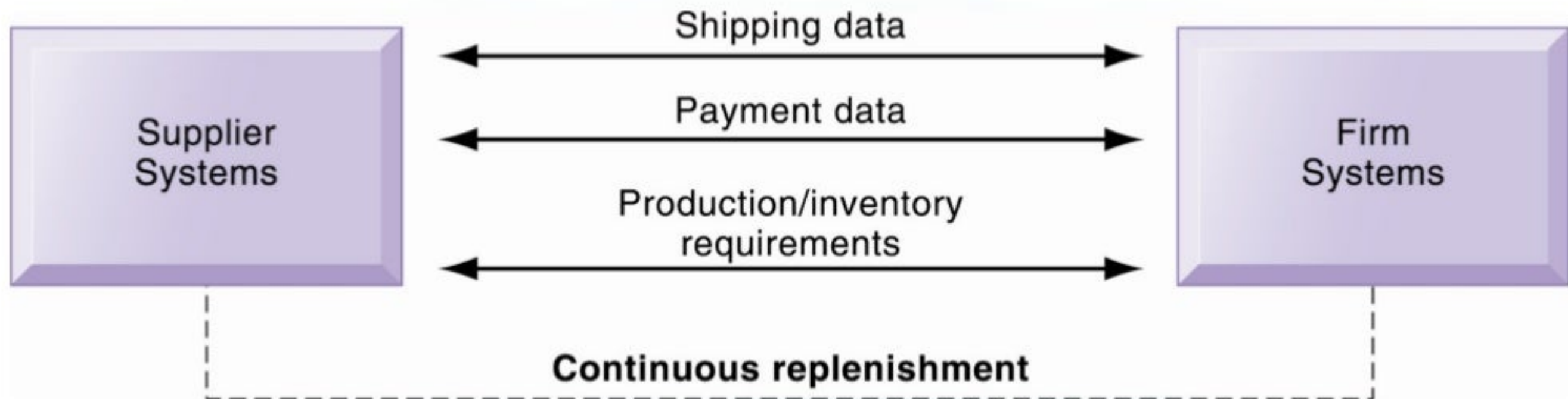


Figure 10-6

Companies use EDI to automate transactions for B2B e-commerce and continuous inventory replenishment. Suppliers can automatically send data about shipments to purchasing firms. The purchasing firms can use EDI to provide production and inventory requirements and payment data to suppliers.



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E-commerce and Business-to-Business Transactions

- **Private industrial network (private exchange)**
 - Large firm using extranet to link to its suppliers, distributors, and other key business partners
 - Owned by buyer
 - Permits sharing of:
 - Product design and development
 - Marketing
 - Production scheduling and inventory management
 - Unstructured communication (graphics and e-mail)



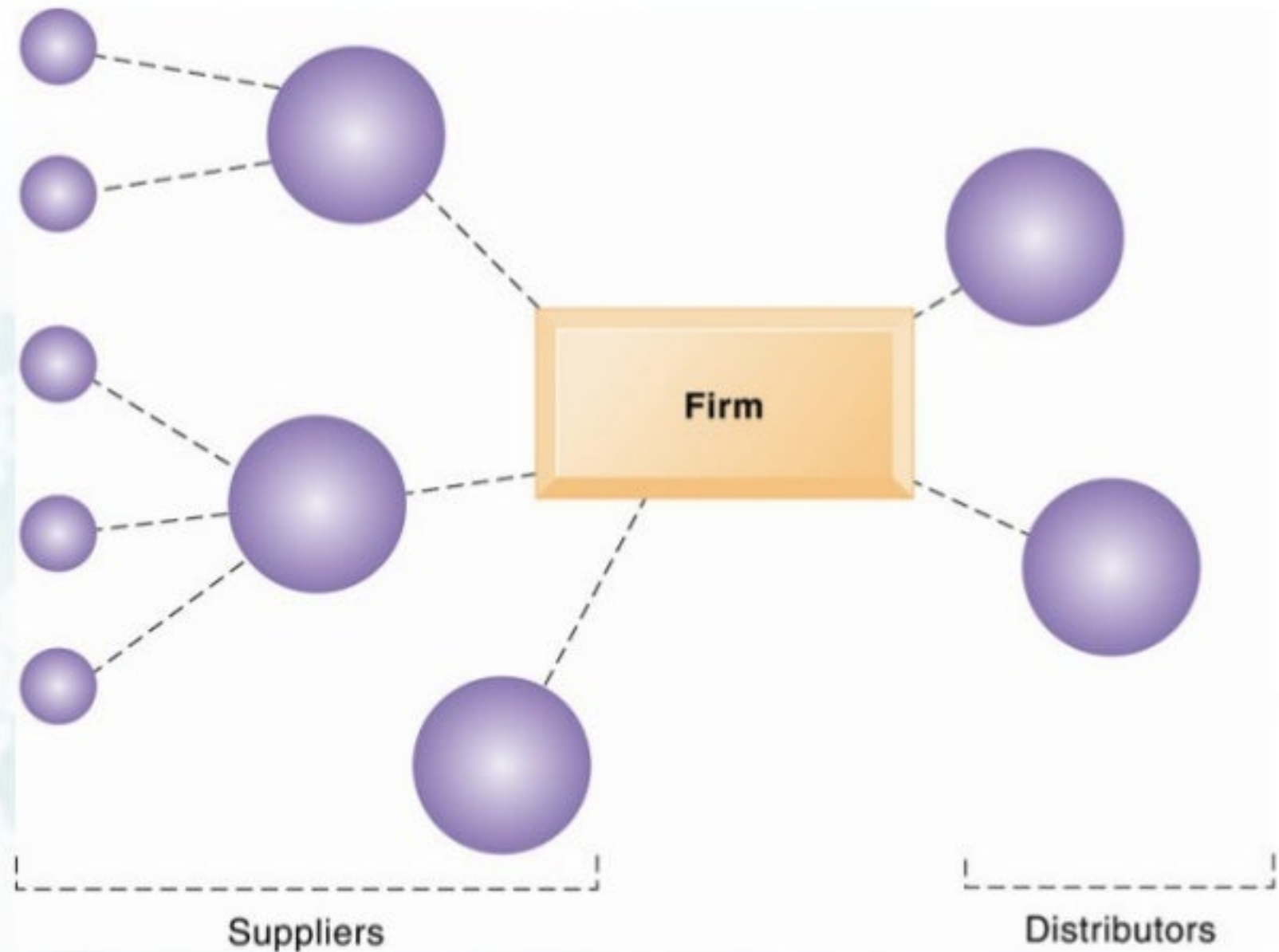
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A Private Industrial Network

A private industrial network, also known as a private exchange, links a firm to its suppliers, distributors, and other key business partners for efficient supply chain management and other collaborative commerce activities.

Figure 10-7





- **Net marketplaces (e-hubs)**
 - Single market for many buyers and sellers
 - Industry-owned or owned by independent intermediary
 - Generate revenue from transaction fees, other services
 - Use prices established through negotiation, auction, RFQs, or fixed prices
 - May focus on direct or indirect goods
 - May be vertical or horizontal marketplaces



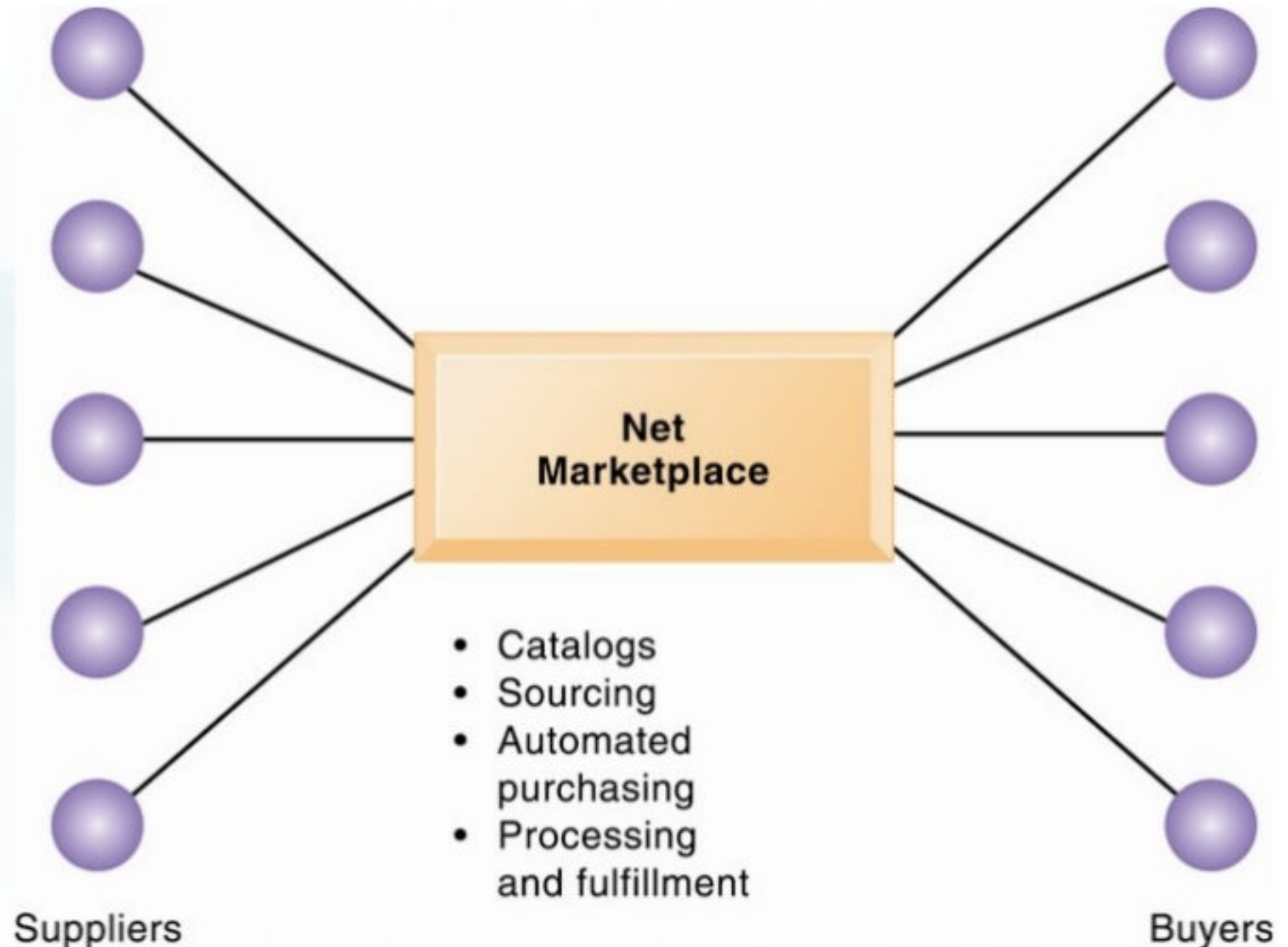
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A Net Marketplace

Net marketplaces are online marketplaces where multiple buyers can purchase from multiple sellers.

Figure 10-8





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E-commerce and Business-to-Business Transactions

- **Exchanges**
 - Independently owned third-party Net marketplaces
 - Connect thousands of suppliers and buyers for spot purchasing
 - Typically provide vertical markets for direct goods for single industry (food, electronics)
 - Proliferated during early years of e-commerce; many have failed
 - Competitive bidding drove prices down and did not offer long-term relationships with buyers or services to make lowering prices worthwhile.



- **M-commerce**
 - In 2014 is 19 percent of all e-commerce
 - Fastest growing form of e-commerce
 - Some areas growing at 50 percent or more
 - Main areas of growth (exclusive of location-based services)
 - Retail sales at top Mobile 400 (Amazon, eBay, etc.)
 - Sales of digital content (music, TV, etc.)



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Consolidated Mobile Commerce Revenues

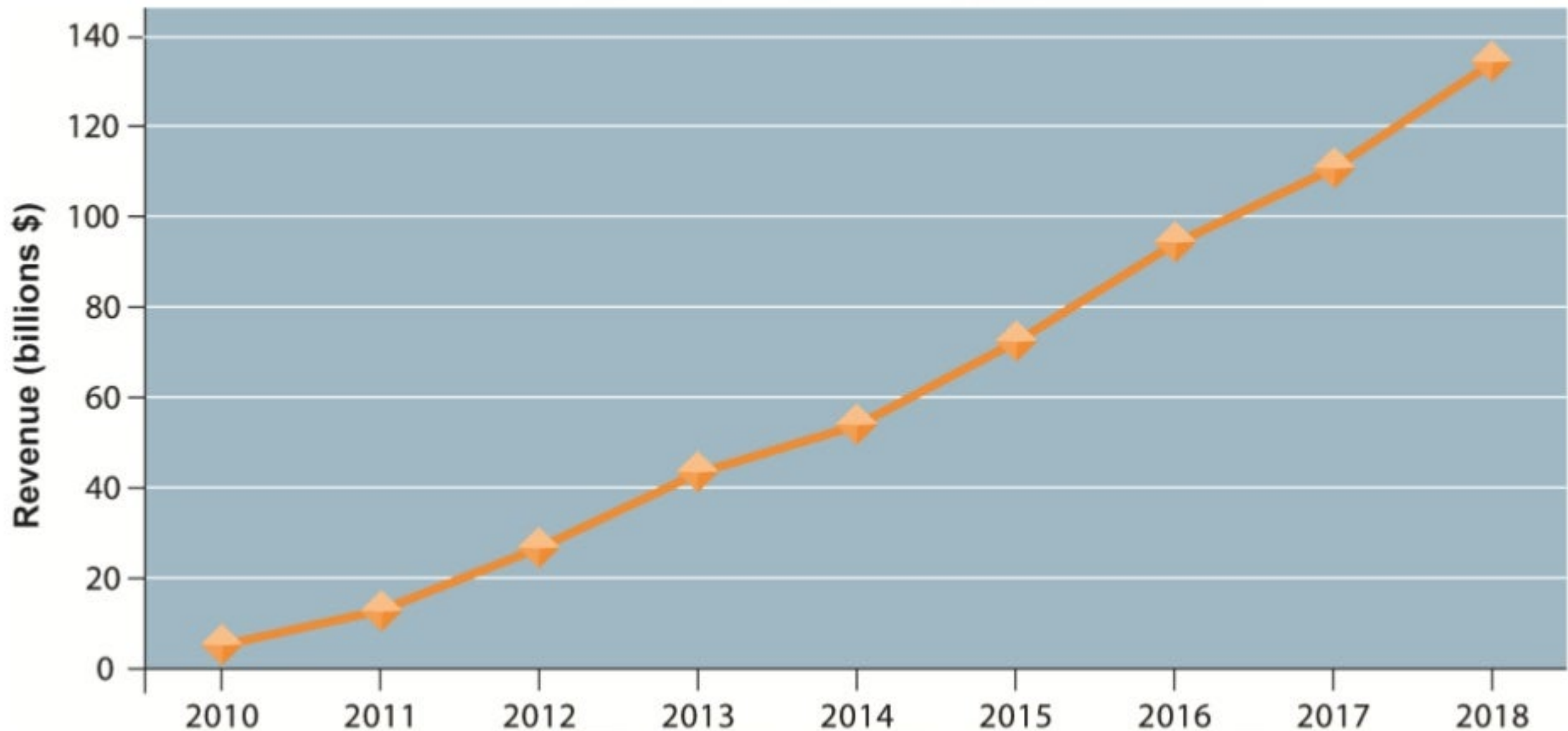


Figure 10-9 Mobile e-commerce is the fastest growing type of B2C e-commerce and represents about 19 percent of all e-commerce in 2014.



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M-commerce and M-commerce Applications

- **Location-based services**
 - **Used by 74 percent of smartphone owners**
 - **Based on GPS map services**
 - **Types**
 - Geosocial services
 - Where friends are
 - Geoadvertising
 - What shops are nearby
 - Geoinformation services
 - Price of house you are passing



Interactive Session: Management

Will Mobile Technology Put Orbitz in the Lead?

Read the Interactive Session and discuss the following questions

- How important is mobile technology in Orbitz's business strategy? Why?
- What management, organization, and technology issues did Orbitz need to address in its mobile strategy?
- Why are mobile phone users much more likely to book same-day hotel room or airline reservations?
- What role does Orbitz for Business play in the company's business strategy?
- How successful is Orbitz's mobile strategy? Explain your answer.



- **Other mobile commerce services**
 - Banks, credit card companies provide account management apps
 - Mobile display advertising
 - iAd, AdMob, Millennial Media, Facebook
 - Ads embedded in games, videos, and mobile apps
 - 55 percent of online retailers have m-commerce Web sites



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Building an E-commerce Presence

- **Most important management challenges**
 - Developing clear understanding of business objectives
 - Knowing how to choose the right technology to achieve those objectives
- **Develop an e-commerce presence map**
 - Four areas: Web sites, e-mail, social media, offline media
- **Develop a timeline: milestones**
 - Breaking a project into discrete phases



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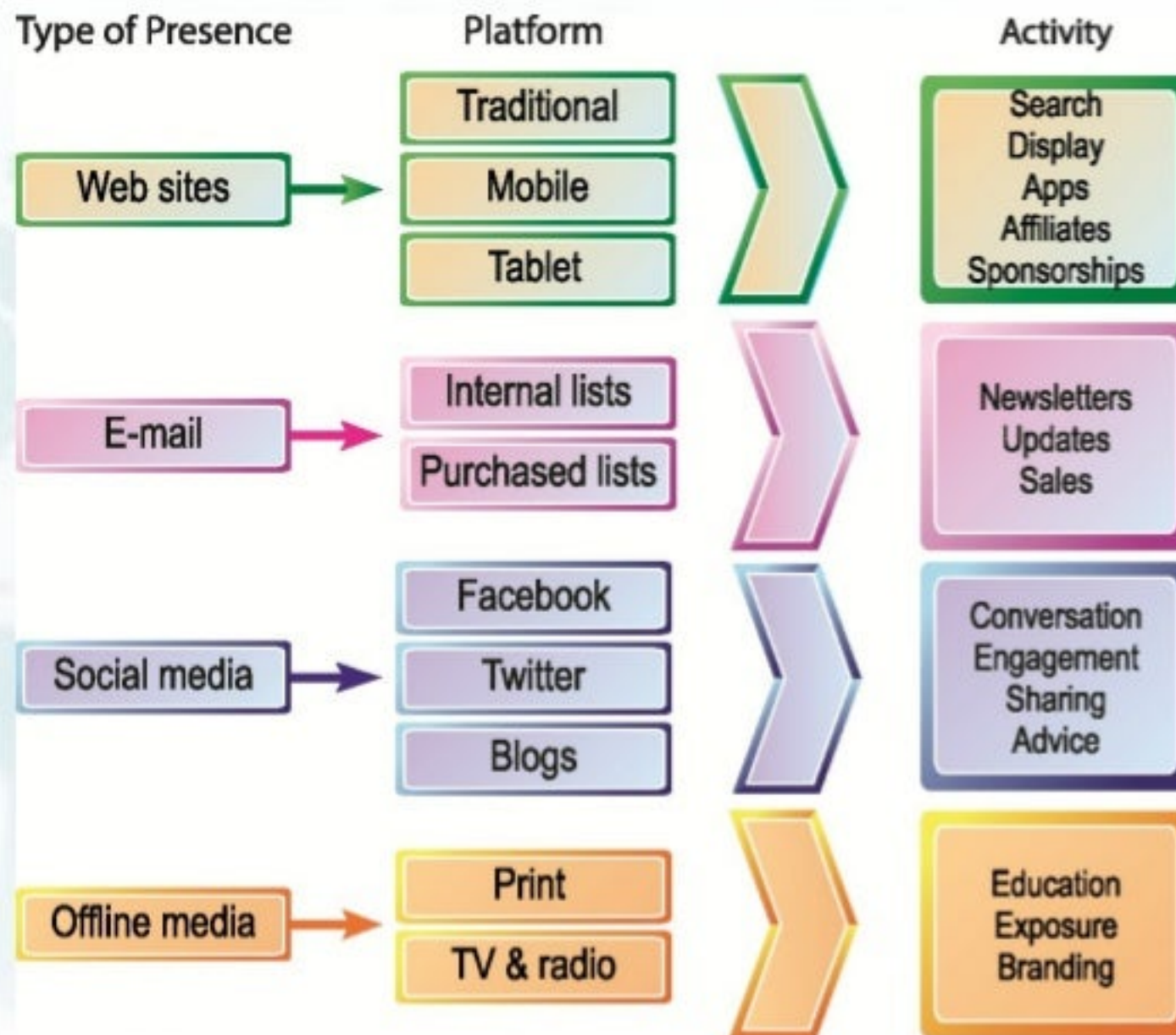
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Building an E-commerce Presence

E-commerce Presence Map

An e-commerce presence requires firms to consider the four different types of presence, with specific platforms and activities associated with each.

Figure 10-10





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Building an E-commerce Presence

E-commerce Presence Timeline

Phase	Activity	Milestone
Phase 1: Planning	Envision Web presence; determine personnel	Web mission statement
Phase 2: Web site development	Acquire content; develop a site design; arrange for hosting the site	Web site plan
Phase 3: Web Implementation	Develop keywords and metatags; focus on search engine optimization; identify potential sponsors	A functional Web site
Phase 4: Social media plan	Identify appropriate social platforms and content for your products and services	A social media plan
Phase 5: Social media implementation	Develop Facebook, Twitter, and Pinterest presence	Functioning social media presence
Phase 6: Mobile plan	Develop a mobile plan; consider options for porting your Web site to smartphones	A mobile media plan

Table 10-8